

FinRobot-NVIDIA

Institutional-Grade Financial AI Agent Platform

Core Technologies

Category	Stack Components
Frontend	Next.js 14, React, Tailwind CSS, TypeScript, Framer Motion
Backend	Python, FastAPI, Uvicorn, Pydantic
Intelligence	NVIDIA NIM (Llama 3 70B Instruct LLM)
Data Feeds	Finnhub, FMP API, NewsAPI, SEC API
Library	Recharts, Lightweight Charts, ReportLab

Real-World Challenges Addressed

- **Information Overload:** Retail traders are overwhelmed by fragmented data from diverse sources.
- **Institutional Gap:** Professional analysis is expensive and slow, creating an 'edge gap' between retail and institutions.
- **Emotional Volatility:** Lack of data-backed reasoning leads to biased trading decisions.

Platform Solutions

- ✓ **Automated Insights:** Uses LLMs to analyze 10-K filings and news instantly.
- ✓ **Structured Reasoning:** Explains the logic behind every forecast using CoT reasoning.
- ✓ **Unified Dashboard:** Real-time price actions meet AI sentiment analysis for cohesive viewing.

Operating Mechanism

The platform runs a multi-agent backend architecture. First, it orchestrates asynchronous calls to global financial APIs (Finnhub, FMP, NewsAPI). This context is injected into the NVIDIA NIM AI engine (Llama 3), which performs a multi-step 'Chain-of-Thought' analysis. The results are streamed to a high-fidelity Next.js dashboard, where interactive charts render both market data and AI insights simultaneously. For on-demand research, the system generates professional-grade PDF equity reports synthesizing hundreds of data points into readable analysis.